

**YEAR 10 SCIENCE - GENERAL  
EARTH AND SPACE SCIENCE  
TEST 1**

NAME: SOLUTIONSMARK: 55

*The universe contains features including galaxies, stars and solar systems and the Big Bang theory can be used to explain the origin of the universe.*

1. Patterns of stars in the night sky are called:

- (a) asteroids.
- (b) galaxies.
- (c) nebulae.
- (d) constellations.

2. A light year is defined as:

- (a) the distance light travels from the closest star.
- (b) the distance from Alpha Centauri to the sun.
- (c) the distance of the Earth from the sun.
- (d) the distance light travels in a year.

3. The main fuel of stars is:

- (a) hydrogen.
- (b) helium.
- (c) carbon.
- (d) gold.

4. Clouds of gas and dust that form spectacular images are:

- (a) stars.
- (b) nebulae.
- (c) planets.
- (d) suns.

5. When a star the size of our sun begins to run out of fuel, it transforms and swells in size until it becomes a:

- (a) white dwarf.
- (b) red giant.
- (c) red dwarf.
- (d) white giant.

6. Which one of the following gives the correct order of size from **smallest to largest**?

- (a) Universe, planet, galaxy, solar system.
- (b) Planet, solar system, galaxy, universe.
- (c) Planet, galaxy, solar system, universe.
- (d) Galaxy, universe, planet, solar system.

| QUESTION | ANSWER |
|----------|--------|
| 1        | D      |
| 2        | D      |
| 3        | A      |
| 4        | B      |
| 5        | B      |
| 6        | B      |
| 7        | C      |
| 8        | A      |
| 9        | D      |
| 10       | C      |
| 11       | D      |
| 12       | A      |
| 13       | A      |
| 14       | B      |
| 15       | D      |
| 16       | D      |
| 17       | B      |
| 18       | B      |
| 19       | B      |
| 20       | C      |
| 21       | C      |
| 22       | B      |
| 23       | C      |
| 24       | C      |
| 25       | B      |

7. The remains of a supernova forms a:
- (a) red giant.
  - (b) brown dwarf.
  - (c) black hole.
  - (d) white dwarf.
8. A galaxy is a group of many billions of stars held together by:
- (a) gravitational forces.
  - (b) electrical forces.
  - (c) magnetic forces.
  - (d) quasars.
9. The Milky Way is most accurately described as a:
- (a) planet.
  - (b) universe.
  - (c) solar system.
  - (d) galaxy.
10. The major source of energy in the Sun is:
- (a) electrical.
  - (b) chemical.
  - (c) nuclear.
  - (d) magnetic.
11. Many scientists expect that Man will travel to other planets in the solar system within the next 100 years. They do not think, however, that Man will travel to planets of other stars in the universe in the next 100 years.
- The main reason that trips to planets of other stars in this period are unlikely is that scientists:
- (a) don't know which stars have planets.
  - (b) fear that the planets may have strange life forms, which could harm the astronauts.
  - (c) doubt that there are any other stars that have planets.
  - (d) realise that such trips would take too long because the distances are so great.
12. A star ten times the size of our Sun would go through the following stages:
- (a) Supernova, neutron star, pulsar, black hole.
  - (b) Pulsar, neutron star, black hole, supernova.
  - (c) Neutron star, pulsar, supernova, black hole.
  - (d) Black hole, supernova, pulsar, neutron star.
13. Most scientists currently favour the theory that the universe:
- (a) once fitted inside a pinhead before "exploding" out into a big bang.
  - (b) has always existed in its current form.
  - (c) began on a large scale, and is slowly shrinking, so that it will one day fit inside a pinhead.
  - (d) has always existed and is continually expanding and contracting.
14. In the sun, hydrogen is transformed into:
- (a) carbon dioxide.
  - (b) helium.
  - (c) oxygen.
  - (d) water.

15. The apparent slow motion of the stars that is observed during the night is due to:
- (a) the orbit of stars around other stars.
  - (b) the orbit of the Earth around the Sun.
  - (c) the effect of parallax.
  - (d) the rotation of the Earth.
16. The observed red shift of some stars is due to their:
- (a) extremely high temperature.
  - (b) extremely low temperature.
  - (c) movement towards the Earth.
  - (d) movement away from Earth.
17. The 'big bang' theory is a possible explanation of:
- (a) the birth of a star.
  - (b) the expansion of the universe.
  - (c) the collapse of the universe.
  - (d) the formation of the solar system.
18. According to the 'big bang' theory, all that existed before the universe began was:
- (a) mass.
  - (b) energy.
  - (c) space.
  - (d) time.
19. Which statement is the best description of the universe?
- (a) A collection of stars, separated by absolutely empty space.
  - (b) A collection of galaxies, nebulae and stars.
  - (c) A collection of solar systems separated by empty space.
  - (d) The Milky Way galaxy, made of stars and planets.
20. The age of the universe is believed by cosmologists to be closest to:
- (a) 15 thousand years.
  - (b) 15 million years.
  - (c) 15 billion years.
  - (d) 15 trillion years.
21. The atoms that came together to form the first stars were pulled together by:
- (a) electric forces.
  - (b) magnetic forces.
  - (c) gravity.
  - (d) friction.
22. **Compared to other stars in the universe**, the mass of the Sun could be best described as:
- (a) huge.
  - (b) average.
  - (c) small.
  - (d) massive.
23. A supernova is the name given to:
- (a) the beginnings of a brand new star.
  - (b) the core of an exploded star.
  - (c) the end stage of a very massive star.
  - (d) the end stage of a very small star.

24. The colour of a star is most directly related to its:

- (a) age.
- (b) location.
- (c) surface temperature.
- (d) volume.

25. Which statement describes evidence that is the basis of the Big Bang Theory?

- (a) All galaxies have been measured to be moving at the same speed.
- (b) Distant galaxies have been measured to be moving away from the Earth faster than closer galaxies.
- (c) All stars have been measured to be moving at the same speed.
- (d) Galaxies are expanding as the stars within the galaxy move away from each other.

**Short Answer**

**Write the answer to the questions in the spaces provided.**

1. Choose the correct definition for the term in the first column of the table below. Write the **correct letter** in the second column.

| TERM          | CORRECT LETTER | DEFINITION                                                      |
|---------------|----------------|-----------------------------------------------------------------|
| Black dwarf   | D              | A: Pattern formed by a group of stars.                          |
| Nova          | F              | B: Formed by the expansion of a star about the size of our Sun. |
| Red giant     | B              | C: Remains of the death of a massive star.                      |
| Black hole    | C              | D: Small, dead star that emits no light.                        |
| Constellation | A              | E: Final stage in the development of a star.                    |
| Protostar     | E              | F: Explosion of the outer layers of an average-size star.       |

(1 mark each)

(6)

2. (a) Name the process occurring in the Sun that produces its energy.

(1)

fusion

(b) What is the name of the major element found in a star?

(1)

hydrogen

(c) Describe what happens to this element in a star for it to produce its energy.

(1)

• Hydrogen becomes helium, releasing energy.

- (d) A star constantly loses mass during its lifetime. Apart from the "pushing" of particles into space as the "solar wind", what happens to this lost mass?  
(Hint - think about Einstein's famous equation.)

(1)

•  $E = mc^2 \rightarrow$  mass is destroyed and becomes energy.

3. Our Sun is yellow in colour. Why is it this colour and not a different one like red or blue?

(2)

- Average temperature = yellow colour. (1)
- Red is cooler, blue is hotter. (1)

4. (a) Our Sun has an interesting pathway until its ultimate "death". Draw a labelled flow diagram to show the stages involved in this pathway.

(2)

Sun  $\rightarrow$  red giant  $\rightarrow$  nova  $\rightarrow$  white dwarf  $\rightarrow$  black dwarf  
[ $\frac{1}{2}$  mark each]

- (b) There are two other possible pathways for stars larger than our Sun. Draw a labelled flow diagram for **ONE** of those pathways.

(2)

Huge  $\rightarrow$  red supergiant  $\rightarrow$  supernova  $\rightarrow$  neutron star  $\rightarrow$  [  $\frac{1}{2}$  mark each ]  
Massive " " " black hole.

5. How is a **black dwarf** different from a **black hole**? Give the characteristics of each.

(2)

- |             |                                                |                            |
|-------------|------------------------------------------------|----------------------------|
| Black dwarf | • Cold + dark<br>• No fusion - no energy.      | [ $\frac{1}{2}$ mark each] |
| Black hole  | • Very dense,<br>• Strong gravitational field. |                            |

6. The diagram below shows the spectrum from our Sun. The same spectrum is seen from all stars.



- (a) Use the diagram to explain what "**red shift**" is. You can draw on the diagram to help your explanation. (2)

- Lines move towards the red end. (1)
- Indicates galaxy is moving away from us. (1)

- (b) Edwin Hubble used "red-shift" to determine the motion of very distant galaxies. List two things that he discovered about these galaxies. (2)

- Furthest galaxies move fastest. (1)
- Galaxies move apart from each other. (1)

- (c) How did this information support the Big Bang Theory? (2)

- Going back in time to "Big Bang", galaxies should be moving apart the quickest. (1)
- Universe is constantly expanding. (1)
- Universe started at one point.

7. (a) Name what was present before the Big Bang took place? What was it made of? (2)

- Singularity (1)
- Pure energy - very dense. (1)

- (b) Was there a "start" to the Big Bang? Explain your answer briefly. (2)

- No (1)
- Unsure of what starts the expansion. (1)

(c) The Big Bang is not an "explosion". Briefly describe what actually takes place.

(2)

- Time and space are created, (1)
- Space expands. (1)

